CHAPTER 3

Limits of functions

S21, S23, S24, S26

Before you begin · Limits of functions

Without a calculator, in six minutes. Answer briefly. Use sign, factorisation and roots. Then choose the targeted follow-up indicated under the questions.

1. Factorise $x^2 - 4$.
2. Give the sign of $\frac{x-1}{x+2}$ on $]-2;1[$.
3. Simplify $\sqrt{x^2}$ for $x < 0$.
4. What is the limit of the sequence $\frac{1}{n+1}$?
5. Determine the dominant term of $3x^3 - x$.
6. Give a forbidden value for $\frac{1}{x-2}$.

Answers

$(x-2)(x+2)$; negative ; $-x$; 0 ; $3x^3$; 2.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Expand (x-2)(x+2) to check. **2.** Make two sign rows, then their quotient. **3.** Test x=-3 then generalise; R28. **4.** Revisit the unit on limits of sequences and the band around 0. **5.** Compare the degrees before any simplification. **6.** Find where the denominator is zero before calculating. The R questions serve as materials for review with the teacher, even if their collective routine is planned for later.

Objectives

Calculate limits at a point and at infinity; resolve an indeterminate form; interpret an asymptote.

Course notes

Meaning of a limit

$f(x) \to \ell$ when $x \to +\infty$ means that $f(x)$ can be made as close as desired to ℓ for all x sufficiently large. For a limit at a, consider x of the domain, close to a and distinct from a. Left and right limits are written $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. When the domain has points on both sides of a, a two-sided limit exists if and only if the two one-sided limits exist and are equal. At an endpoint of the domain, we study the limit on the only accessible side.

Graphical interpretation

If $f(x) \to \ell \in \mathbb{R}$ when $x \to +\infty$ or $-\infty$, the line $y = \ell$ is a horizontal asymptote in this direction. If $f(x) \to \pm\infty$ when $x \to a$ on one side, the line $x = a$ is a vertical asymptote.

A curve can cross a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

References and operations

At $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. At $-\infty$, the sign of x^m depends on the parity of m. The rules for sum, product, quotient and comparison are those for sequences, with the same indeterminate forms.

Example. For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. The limits at $\pm\infty$ are equal to 2. At 3^-, $f \to -\infty$; at 3^+, $f \to +\infty$. The asymptotes are $x = 3$ and $y = 2$.

Conjugate. For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, be careful: $\sqrt{x^2} = |x| = -x$.

Exponential and comparative growth

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

We deduce $x^m e^{-x} \to 0$ at $+\infty$.

Proof of the limits of the exponential. The study of $e^x - x - 1$ gives $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ at $+\infty$. Setting $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

Proof of a comparative growth result. For every integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If the inequality holds at order k, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value at 0 is positive. The inductive step follows. By choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

Choose a transformation

Quotient of polynomials: factor out the highest powers. Difference of square roots: multiply by the conjugate. Quotient near a zero: factorise and study the signs. Exponential versus power: comparative growth, after a change of variable if necessary.

Comparison and composition: conditions of use

If $g(x) \leq f(x) \leq h(x)$ near the point or infinity studied, and if g, h have the same real limit ℓ, then $f(x) \to \ell$ (squeeze theorem). If $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit $-\infty$, we use an upper bound by a function tending to $-\infty$. If $u(x) \to a$ and v is continuous at a, then $v(u(x)) \to v(a)$, when the composite is defined. For example $\sqrt{1+\frac{1}{x}} \to 1$ at $+\infty$. For an infinite limit of the outer function, check from which side $u(x)$ approaches the bound: $\frac{1}{u(x)}$ does not have the same behaviour according to the sign of $u(x)$.

Read a limit, choose a transformation

3.1 The approach side is part of the question

First use: S23.

Worked example · A vertical asymptote

For $f(x) = 1 + \frac{1}{x-2}$, the domain is $\mathbb{R} \setminus \{2\}$. When $x \to 2^-$, the denominator is negative and tends to 0, so $f(x) \to -\infty$. When $x \to 2^+$, $f(x) \to +\infty$. The line $x = 2$ is a vertical asymptote; no value $f(2)$ exists. At infinity, $f(x) \to 1$.

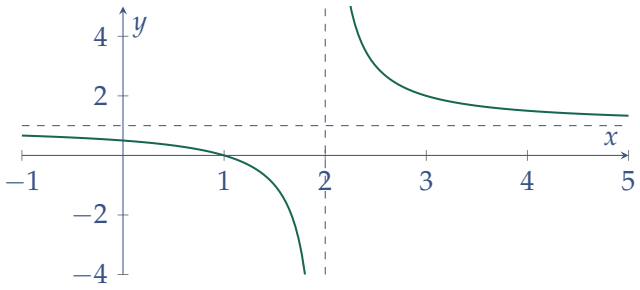


Figure 3.1. Find the two one-sided limits on the drawing: the branches are not joined.

A1 · Immediate application Determine the limits of $g(x) = 3 - \frac{2}{x+1}$ at -1^-, -1^+, $-\infty$ & $+\infty$. Give the asymptotes.

3.2 An indeterminate form calls for a calculation

First use: S24.

Worked example · The sign hidden in a square root

For $x < 0$, $\sqrt{x^2} = |x| = -x$. Thus, at $-\infty$,

$$\frac{\sqrt{x^2+1}}{x} = \frac{|x|\sqrt{1+1/x^2}}{x} = -\sqrt{1+1/x^2} \longrightarrow -1.$$

At $+\infty$, the same expression tends to 1. Omitting the absolute value would produce a positive limit for an expression that is actually negative.

A2 · Fading guidance Calculate at $+\infty$ the limits of $\frac{2x^2+1}{x^2-x}$, $\sqrt{x^2+4x} - x$ and $x\left(\sqrt{1+1/x} - 1\right)$. For the first, we suggest dividing by x^2 ; choose freely for the others.

A3 · Construct examples Construct three quotients whose limit has the form $0/0$ at 0, whose limits are respectively 0, 1 and $+\infty$. State the side of approach if necessary.

3.3 Comparative growth and change of variable

First use: S26; logarithm after S49.

Property · Exponential and powers · results from the lesson

For every integer $m \geq 1$, $e^x/x^m \to +\infty$ & $x^m e^{-x} \to 0$ at $+\infty$. To transfer these results, identify the new variable and its limit.

Worked example · Transferring a result

When $x \to -\infty$, set $t = -2x$ gives $t \to +\infty$ & $x^2e^{2x} = \frac{1}{4}t^2e^{-t} \to 0$. After S49, the substitution $t = 1/x$ in $x \ln x$ for $x \to 0^+$ gives $-(\ln t)/t \to 0$. The logarithm is not used before it has been studied.

A4 · Deferred application After S26, calculate $\lim_{x\to+\infty}(x^2+3x)e^{-x}$ & $\lim_{x\to-\infty} xe^x$. Say why formally replacing x by infinity is not a justification.

3.4 An asymptote may be crossed

First use: S24.

Definition · Horizontal Asymptote

If $f(x) - \ell \to 0$ at one of the infinities, $y = \ell$ is an asymptote at this infinity. This definition requires a vertical distance that tends to 0; it does not rule out any intersection at a finite distance.

Worked example · Two behaviours

On $\mathbb{R}$, $p(x) = 1 + 1/(1+x^2)$ remains strictly above $y = 1$. The function $q(x) = 1 + x/(1+x^2)$ crosses this same line at $x = 0$, while tending to 1 at both infinities. For q, the sign of $q(x) - 1$ is that of x.

A5 · Refute and explain A student says that the curve of q cannot cross its asymptote. Give the coordinates of the point of intersection and the signs on either side. Is this consistent with the two limits?

A6 · Choice of strategy Determine a, b so that $h(x) = \frac{ax+b}{x-1}$ has the horizontal asymptote $y = 2$ and satisfies $h(0) = 3$. Then study the one-sided limits at 1 and the position of the curve relative to $y = 2$.

One minute, spoken aloud. For each of the expressions in the "Methods" exercise, first state what prevents a conclusion from the operations on limits, then explain the role of the transformation.

Delayed review · S49 then S81. Redo the change of variable with $\ln x / x$; sort again the limits, mixing logarithms, exponentials and roots.

Investigation

A curve close to a line - 30 min.

Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. Let $\mathcal{C}_f$ its curve in a coordinate system of the plane.

1. Show that, for every $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.

2. Determine the limits of f at $-\infty$, at 1^-, at 1^+ and at $+\infty$.

3. Study the sign of $f(x) - 3$, then specify the position of $\mathcal{C}_f$ with respect to the line with equation $y = 3$.

4. Determine a real number $A > 1$ such that, for every $x > A$, $|f(x) - 3| < 0{,}01$.

5. Does curve $\mathcal{C}_f$ intersect its horizontal asymptote? Justify and explain why an overly wide graphing window can give a misleading impression.

Progressive exercises

E1 · Direct application. Determine $\lim\limits_{x \to +\infty} \dfrac{5x^3 - x}{2x^3 + 7}$ justifying.

E2 · Application and justification. Let f the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits at 2^-, at 2^+, at $-\infty$ and at $+\infty$.

E3 · Method to organise. Determine $\lim\limits_{x \to 1} \dfrac{x^2 - 1}{x - 1}$ specifying why the simplification used is valid near 1.

E4 · Choice of strategy. Determine $\lim\limits_{x \to +\infty} x^3 e^{-2x}$. State the comparative growth result used.

E5 · Synthesis or proof. Determine $\lim\limits_{x \to -\infty} \left(\sqrt{x^2 + 1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.

Work from the textbook or a past paper

Barbazo, p. 96, Exercises 26 and 29. In S23: exercise 29, to link limits and asymptotes. At home after S24: exercise 26, to construct three counterexamples to a faulty calculation rule on limits.

Homework

DM3 - A family of curves - 45 min.

For every real number a, consider the function f_a defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. Let $\mathcal{C}_a$ be its graph.

1. Determine the domain of f_a, then its limits at $-\infty$ and at $+\infty$.

2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of a, state the position of $\mathcal{C}_a$ with respect to the line with equation $y = a$.

3. Study the variations of f_a distinguishing the cases $a < 1$, $a = 1$ and $a > 1$. State any extrema and where they are attained.

4. Take $a = 3$. Solve, for $x \geq 0$, the inequality $|f_3(x) - 3| \leq 0{,}02$.

MCQ A · one-sided limits and transformations

After S24; MCQ B after S26.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $\lim_{x\to 2^-} \frac{1}{x-2}$ is:

A. $+\infty$ B. $-\infty$

C. 0 D. 2

2. For $x < 0$, $\sqrt{x^2}$ is:

A. x B. x^2

C. $-x$ D. $\pm x$

3. $\lim_{x\to+\infty} \frac{3x^2-x}{x^2+4}$ is:

A. 0 B. $+\infty$

C. -1 D. 3

4. $\lim_{x\to 1} \frac{x^2-1}{x-1}$ is:

A. 2 B. 0

C. $+\infty$ D. does not exist

5. For $f(x) = 1 + \frac{\sin x}{x}$, $x > 0$, the line $y = 1$:

A. is not an asymptote B. is an asymptote with no intersection

C. is an asymptote crossed infinitely often D. is tangent at every point

MCQ B · comparative growth and choice of tool

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. $\lim_{x\to+\infty} x^2 e^{-x}$ is:

A. $+\infty$ B. 0

C. 1 D. $-\infty$

7. $\lim_{x\to-\infty} xe^x$ is:

A. 0 B. $+\infty$

C. $-\infty$ D. 1

8. The appropriate tool for $\sqrt{x^2+2x} - x$ at $+\infty$ is:

A. the binomial B. the IVT

C. differentiation D. the conjugate

9. If $f(x) \to 0$ and $g(x) \to +\infty$the product:

A. always tends to 0 B. always tends to $+\infty$

C. needs further study D. is impossible

10. If $f(x) \geq x^2$ for $x \geq 1$, then:

A. $f(x)/x^2 \to 1$ B. $f(x) \to +\infty$

C. f is increasing D. $f(x) - x^2 \to 0$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. A graph never crosses its horizontal asymptote.

2. A finite limit at a requires that $f(a)$ is defined.

3. The symbol ∞/∞ gives a limit equal to 1.

MCQ solutions

1. **B.** The denominator tends to 0 through negative values. The side of approach determines the sign.

2. **C.** The square root is positive or zero; $\sqrt{x^2} = |x| = -x$ for $x < 0$. This is not an equation with two solutions.

3. **D.** Dividing by x^2 gives $(3 - 1/x)/(1 + 4/x^2) \to 3$.

4. **A.** For $x \neq 1$, the quotient is $x + 1$. An excluded value does not make a limit impossible.

5. **C.** The difference tends to 0 and is zero at $k\pi$, $k \geq 1$, with a change of sign. An asymptote can be crossed.

6. **B.** This is x^2/e^x, which tends to 0 by comparison of growth rates. The exponential dominates every fixed power.

7. **A.** Set $t = -x \to +\infty$: the expression is $-t/e^t \to 0$.

8. **D.** Multiplying by the conjugate gives 2/ $(\sqrt{1 + 2/x} + 1) \to 1$.

9. **C.** With $g = x$, take $f = 1/x^2$, $1/x$ or 1/ $\sqrt{x}$: we obtain respectively 0, 1 and $+\infty$.

10. **B.** The comparison theorem is enough for the limit. 2 x^2 disproves both limiting equalities; the inequality does not give monotonicity.

True/false. False: question 5. False: question 4. False: $x/x \to 1$, $x/x^2 \to 0$, $x^2/x \to +\infty$.

Solutions to the applications and guidance

Application 1. At -1^- : $+\infty$; at -1^+ : $-\infty$; at both infinities: 3. Asymptotes $x = -1$ and $y = 3$.

Application 2. Limits: 2; $\frac{4x}{\sqrt{x^2+4x}+x} = \frac{4}{\sqrt{1+4/x}+1} \to 2$; $\frac{1}{\sqrt{1+1/x}+1} \to 1/2$. The expressions are defined for x large enough.

Application 3. For example $x^2/x \to 0$, $x/x \to 1$, $x/x^2 = 1/x \to +\infty$ at 0^+. The functions are considered outside 0.

Application 4. The first tends to 0 by linearity and comparative growth. For the second, $t = -x \to +\infty$ gives $xe^x = -te^{-t} \to 0$. The symbol ∞ is not a real number at which we evaluate a function.

Application 5. $q(0) = 1$. $q(x) - 1$ is negative for $x < 0$, positive for $x > 0$. The limits concern the infinite ends of the domain, so they do not contradict this intersection.

Application 6. The limit at infinity is a, therefore $a = 2$; $h(0) = -b = 3$, so $b = -3$. Thus $h(x) = 2 - 1/(x-1)$: limits $+\infty$ at 1^-, $-\infty$ at 1^+ ; curve above 2 to the left of 1 and below on the right.

Reading then calculation

Before any table, have students anticipate the sign of the quotient. A numerical result contradicting the expected sign should trigger a check. Have students draw p and q in the same window to challenge the image of an "impassable barrier".

Teaching notes and solutions

Guidance for the teacher

In S21, link each limit to a reading of the graph. In S23, require the signs of the denominator on the left and right. In S24, have students choose the transformation before calculating: factorisation, conjugate, local expansion not used. In S26, separate the proof of the comparative growth results from their use. Oblique asymptotes are not required here.

Solutions to the activity

1. Since $3 \quad x-2=3(x-1)+1$, for $\quad x \neq 1, f(x)=3+\frac{1}{x-1}$.

2. At $\pm\infty, \frac{1}{x-1} \to 0$: the limits equal 3. At $1^{-}, x-1 \to 0^{-}$, therefore $f(x) \to -\infty$; at $1^{+}, x-1 \to 0^{+}$, so $f(x) \to +\infty$.

3. The sign of $f(x)-3=\frac{1}{x-1}$ is negative for $x<1$, positive for $x>1$. The curve is therefore respectively below and above the asymptote $y=3$.

4. For $x>1, |f(x)-3|<0.01$ is equivalent to $\frac{1}{x-1}<0.01$, then $x>101$. We can choose $A=101$.

5. No: $f(x)-3=\frac{1}{x-1}$ is never zero. The curve may appear to merge with the asymptote at the precision of the screen, but it does not meet it.

E1. For $x \neq 0$, the quotient equals $\frac{5-\frac{1}{x^2}}{2+\frac{7}{x^3}}$. At $+\infty$, the reciprocal powers tend to 0 and the denominator to $2 \neq 0$. The limit is $\frac{5}{2}$.

E2. At 2, $(x-2)^2 \to 0$ through strictly positive values on both sides; its reciprocal therefore tends to $+\infty$. At $\pm\infty$, the square tends to $+\infty$ and its reciprocal to 0.

E3. For $x \neq 1, x^2-1=(x-1)(x+1)$, so the quotient equals $x+1$. Its limit at 1 is therefore 2. This simplification computes the limit; it does not make the original function defined at 1.

E4. Let $t=2x \to +\infty$. Then $x^3e^{-2x}=\frac{t^3}{8e^t} \to 0$, by comparative growth of the exponential and the power.

E5. The conjugate gives $\sqrt{x^2+1}+x=\frac{1}{\sqrt{x^2+1}-x}$. When $x \to -\infty$, both terms $\sqrt{x^2+1}$ and $-x$ tend to $+\infty$; the denominator therefore tends to $+\infty$. The limit is 0, approached from above.

Solutions to homework DM3

1. Since $x^2+1>0$ on $\mathbb{R}$, the domain is $\mathbb{R}$. Dividing by x^2 for $x \neq 0, f_a(x)=\frac{a+\frac{1}{x^2}}{1+\frac{1}{x^2}} \to a$ at $\pm\infty$.

2. $f_a(x)-a=\frac{ax^2+1-a(x^2+1)}{x^2+1}=\frac{1-a}{x^2+1}$. For $a<1$, the curve is above $y=a$; for $a>1$, it is below; for $a=1$, it coincides with this line.

3. By the quotient rule, $f_a'(x)=\frac{[2ax(x^2+1)-2x(ax^2+1)]}{(x^2+1)^2}=\frac{2x(a-1}{(x^2+1)^2}$. If $a>1$, the sign is that of x : decreasing on $]-\infty,0]$, increasing on $[0,+\infty[$; minimum $f_a(0)=1$ and no maximum (the bound a is not reached). If $a<1$, the variations are reversed: maximum 1 and no minimum. For $a=1$, the function is constant, with value 1.

4. For $a=3$ and $x \geq 0$, the inequality becomes $\frac{2}{x^2+1} \leq 0{,}02$. The denominator is positive, so it is equivalent to $x^2+1 \geq 100$, i.e. $x \geq \overline{99}$.

Solutions to the textbook or past-paper work

Exercise 26. When $x \to +\infty$, the functions $f(x) = x$ and $g(x) = x$ both tend to $+\infty$, but their difference is zero. With $f(x) = 2x$, $g(x) = x$, the difference tends to $+\infty$; by swapping f and g, it tends to $-\infty$. These examples show that the difference is an indeterminate form.

Exercise 29. The denominator is zero at $\frac{3}{2}$, therefore $D_f = \mathbb{R} \setminus \{\frac{3}{2}\}$. We write

$$f(x) = \frac{5 - 2x}{2x - 3} = -1 + \frac{2}{2x - 3}.$$

Thus $f(x) \to -1$ at both infinities. At $\frac{3}{2}$, the numerator of the second term is positive and its denominator tends to 0: the left-hand limit is $-\infty$, the one on the right is $+\infty$. The asymptotes have equations $y = -1$ and $x = \frac{3}{2}$.